Information Signaling in Artist Branding: Consumer Utility and Simulated Popularity

Ruining Li
Business Economics
York University
June, 2026

1. Introduction

In the digital era, people rely less on CDs than they did previously, but also on a range of digital audio and media players and content delivery methods. The music company developed artist brands that could be monetized through records, concerts, and traditional merchandise, such as T-shirts and souvenirs. (Meier and Leslie M, 2017). According to Lieb and Kristin J. (2018), drawing on Kotler, P. and Armstrong, G. (2011), branding has traditionally been defined as a means of identifying and differentiating products. Lieb and Kristin J. (2018) also discuss the definition of branding from Keinan, A., and Avery, J. (2008); branding is more likely to be perceived by consumers. Although artist branding may influence consumers in multiple ways, information signaling represents only one of the possible mechanisms. This project does not assume information signaling is the dominant mechanism. Instead, it isolates this channel to develop a simplified theoretical model to investigate how branding may influence consumers' choice through information signaling.

2. Literature Review

2.1 The emergence of branding

Following Meier and Leslie (2016), this section reviews the emergence of artist branding in the digital music industry. During the digital era, people rely less on physical CDs; music audiences have many ways to access abundant musical content, which are a range of digital audio/media players and content delivery methods. For instance, smartphones, MP3 players, tablets, digital downloads, and music streaming services. The sharp decline in CD sales in the digital era propelled the music industry reconfiguration of popular music's core commodity form; they suggest "artist-brands" have become a basis of capitalization. Meier and Leslie (2016) note that the music industry shifted from manufacturing to rights management, citing Frith (1988). This transition suggests that artist branding is no longer limited to promoting recorded music, but has become closely tied to commercial rights management.

2.2 Branding as an information signal

Among the various functions of artist branding, this study focuses especially on its information signaling function. Lieb and Kristin J. (2018) cite Kotler and Armstrong (2011) in defining brands, that brand is a "name, term, sign, symbol, or design, or a combination of them, intended to identify the goods and services of one seller or a group of sellers and differentiate them from those of competitors". Branding is used to show information about a product to consumers and differentiate it from competitors, which may influence consumer choice in different ways. We will discuss the mechanism of information signaling by using the food industry in the following section.

2.3 Consumer interpretation of branding information

Using the food industry as an example, Caswell and Mojduszka (1996), drawing on Hooker and Caswell (1996), food product quality attributes include food safety, nutritional value, packaging, and process attributes. Consumers are willing to pay more for food products that provide additional safety because greater safety can increase consumer utility. However, based on Swinbank (1993), Caswell and Mojduszka suggested that consumers are willing to pay less for each additional unit of safety, which indicates diminishing marginal utility from additional food safety. According to Van Ravenswaay (1995), household utility is not only affected by consuming goods, but also by their health conditions. Recovering their health will increase current and future utility. Since consumers cannot observe quality directly under information asymmetry, their decisions may depend on perceived quality rather than actual quality. Caswell and Mojduszka argue that if a consumer has an incorrect perception of the quality attribute of foods, they may lose utility. For instance, a consumer may spend more on a higher-than-optimal level of food safety or take more risks than they would.

Although these findings are derived from the food industry, the underlying mechanism can be generalized to branding information more broadly. Within information asymmetry, information presented on products can influence consumer utility by shaping their perceptions of product quality. In other words, information presented on products may change their purchase behaviour. As discussed in the previous section, branding is also defined as a way to convey information. Therefore, branding may influence consumer behaviour theoretically.

However, information is not the only mechanism through which branding affects consumers. Branding may also influence consumer perception through accumulated experience and reputation.

3. Building Model

We assume diminishing marginal benefits of information presented on music branding to consumers:

$$\boldsymbol{U = \log(1 + S) + \varepsilon}$$

U is the utility function of each listener with 1 observable explanatory variable S, and one stochastic error term ε. S stands for the strength/intensity of information signal, $S \in [0, \infty)$; $\varepsilon \sim N(0, 1)$, which included the channels except information signaling (e.g. reputation, personal preference, social influence).

$$\frac{dU}{dS} = \frac{1}{(1+S)} > 0$$

$$\frac{d^2U}{dS^2} = -\frac{1}{(1+S)^2} < 0$$

Therefore, the utility function is strictly concave and exhibits a diminishing marginal benefit of information signaling intensity.

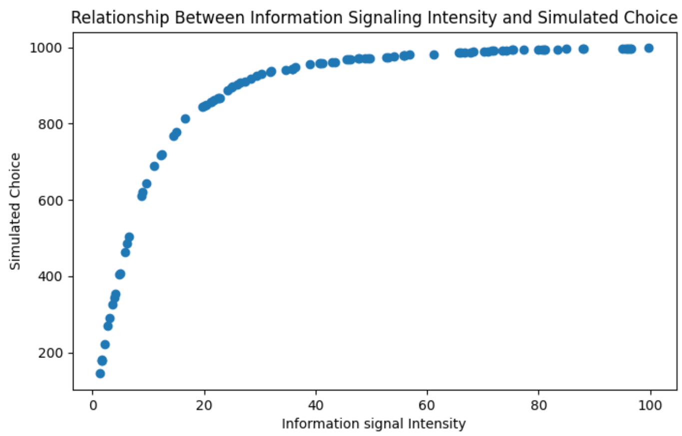
4. Methodology

The simulation generates 100 music brands with uniformly distributed information signal intensity in the range [0,100) and 1000 listener utility functions, which $\varepsilon \sim N(0, 1)$. Each brand was evaluated by one listener; repeat this process until 1000 listeners finished the evaluation. If the utility is greater than 2, we recorded the listener attracted by branding, which is reflected on scatter plot.

To further examine the relationship between information signaling in branding and utility, an additional analysis was conducted. Instead of simulated listener choices and play counts, I focused on plotting graphs of information signal intensity (S) against utility function (U); two scenarios were considered: $\varepsilon \sim N(0, 1)$ and $\varepsilon \sim N(0, 0.1^2)$, to illustrate the effect of random variation on the observed relationship.

5. Results

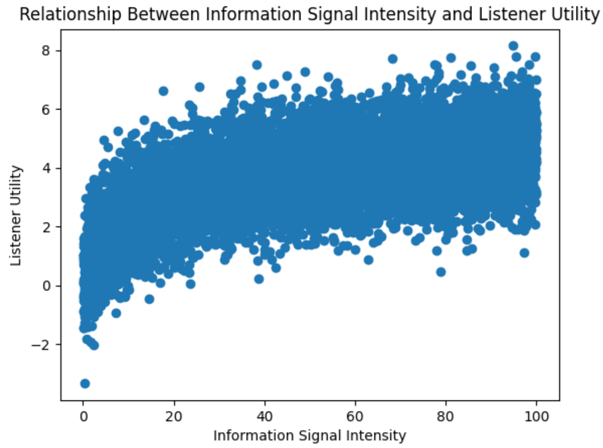
Figure 1 shows the relationship between information signal intensity and simulated choice. The results indicate that the marginal impact of information signaling diminishes as the intensity of the information signal increases. At low levels of intensity, simulated choice increases rapidly. However, the growth slows as intensity becomes stronger.



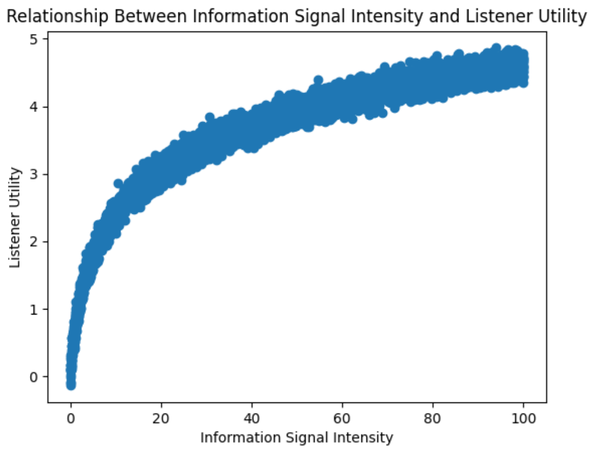
(Figure 1)

6. Additional Analysis

Additional analysis will focus on the relationship between information signaling and listener utility. To clearly illustrate the relationship, the large variation in Figure 2 obscures the relationship between information signaling and utility, which has a standard deviation of 1. Figure 3 shows the same relationship with a standard deviation of 0.1. Therefore, due to the smaller standard deviation, the relationship between information signaling and utility becomes more apparent in Figure 3.



(Figure 2)



(Figure 3)

7. Results for Additional Analysis

Figure 3 shows that the assumed diminishing relationship becomes more apparent when random variation is small. In contrast, Figure 2 illustrates that greater variation may obscure the relationship between information signaling and consumer utility. This suggests the apparent influence of information signaling is sensitive to the magnitude of other factors in the stochastic error term.

8. Discussion

The results illustrated that information signaling appears to be more influential when initial information intensity is low; during information asymmetry, consumers may respond to stronger information signals. For music producers, information signaling may be useful for attracting listeners, but it cannot guarantee unlimited growth. The impact of information in branding may vary depending on external factors. Music producers should not only focus on labeling information in their brand, but also consider external effects.

9. Limitations

The model of information signaling and consumer choice does have many limitations; the impact of information signaling may not be the dominant mechanism; music quality, public tastes, artist's reputation, and social influence could affect the results. According to additional analysis, with a large random shock, the impact of information in branding will be dispersed. Therefore, the model should be interpreted as a stylized representation of information signaling under controlled assumptions, rather than a complete representation of real-world music consumption.

Another limitation of this project is that the relationship between information signal intensity and listener choice is already determined by the utility function. The logarithm in the utility function describes diminishing marginal benefits of information signal in branding. However, different utility functions may have different results, for instance, linear or quadratic functions. Therefore, the results should be interpreted as a prediction in a specific situation rather than real-world phenomenon.

10. Conclusion

This project investigated the relationship between information signal in branding and listener choice through a simplified simulated model. Additional analysis also suggested that the impact of information labeling is sensitive to random shocks.

The results suggested that stronger information signal intensity increases simulated choice. However, the marginal effect diminishes as information signal intensity increases. For music producers, information signaling might be useful to attract listeners, but not forever because they also have to consider other factors.

11. References

Caswell, J. A., & Mojduszka, E. M. (1996). Using Informational Labeling to Influence the Market for Quality in Food Products. American Journal of Agricultural Economics, 78(5), 1248–1253. https://doi.org/10.2307/1243501

Lieb, K. J. (2013). Gender, Branding, and the Modern Music Industry: The Social Construction of Female Popular Music Stars. Taylor and Francis. https://doi.org/10.4324/9780203071786

Meier, L. M. (2017). Popular music as promotion: music and branding in the digital age.

van Ravenswaay, E. O., & Eileen O. van Ravenswaay. (1995). Valuing Food Safety and Nutrition: The Research Needs. In J. A. Caswell (Ed.), Valuing Food Safety and Nutrition (1st ed., pp. 3–26). Routledge.